Clarifying the Cross Product in Craig's Velocity Formula

In Craig's robotics book, the recursive velocity formula includes the term:

$$v_{i+1} = v_i + \omega_i \times P_{i+1}$$

At first glance, ω × P may seem confusing if read as a direct multiplication of two 3×1 vectors. In fact, it is the standard 3D vector cross product. Below are two equivalent ways to write it.

1. Cross Product in Determinant Form

For two vectors $\omega = [\omega_x, \omega_y, \omega_z]^T$ and $P = [P_x, P_y, P_z]^T$, the cross product is:

$$\omega \times P = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \omega_x & \omega_y & \omega_z \\ P_x & P_y & P_z \end{vmatrix}$$

Expanding this determinant gives:

$$\omega \times P = (\omega_y P_z - \omega_z P_y)\hat{\imath} + (\omega_x P_z - \omega_z P_x)\hat{\jmath} + (\omega_x P_y - \omega_y P_x)\hat{k}$$

2. Skew-Symmetric Matrix Form

Alternatively, the cross product can be represented using the skew-symmetric matrix of ω:

$$[\omega] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Then the cross product is simply:

ω × P = [ω] P

☑ Takeaway

Craig's ω × P is just the ordinary vector cross product. It can be written either in determinant form (as we first learn in vector calculus) or in skew-symmetric matrix form (which is more convenient in robotics derivations).